Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

JOINS AND SET OPERATIONS

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Section – 3

CROSS JOIN

Create database:

```
CREATE DATABASE IF NOT EXISTS bookflow2_db;
```

Output:

```
113  23:09:11  CREATE DATABASE IF NOT EXISTS bookflow2_db
```

Use database:

```
USE bookflow2_db;
```

Output:

```
114  23:10:42  USE bookflow2_db
```

Create Tables

```
CREATE TABLE class1 (
    id INT,
    name VARCHAR(30)
);
CREATE TABLE class_info (
    id INT,
    address VARCHAR(30)
);
```

Output:

```
115  23:11:37  CREATE TABLE class (  id INT,  name VARCHAR(30) )
116  23:11:39  CREATE TABLE class_info (  id INT,  address VARCHAR(30) )
```

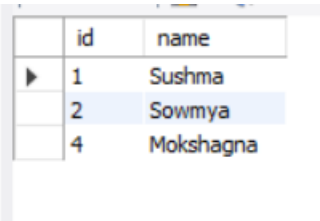
Insert Data:

```
INSERT INTO class VALUES
(1,'Sushma'),
(2,'Sowmya'),
(4,'Mokshagna');
INSERT INTO class_info VALUES
(1,'DELHI'),
(2,'MUMBAI'),
(3,'CHENNAI');
```

Show data:

```
SELECT * FROM class;
```

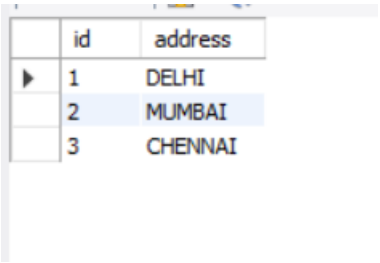
Output:



id	name
1	Sushma
2	Sowmya
4	Mokshagna

```
SELECT * FROM class_info;
```

Output:

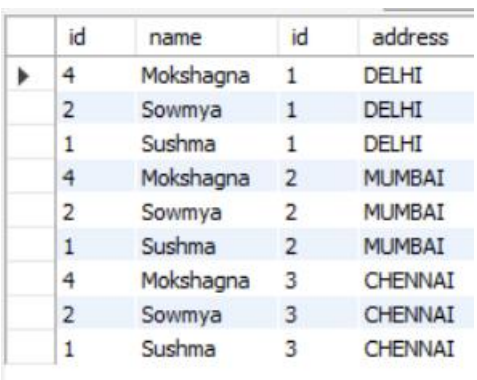


id	address
1	DELHI
2	MUMBAI
3	CHENNAI

Cross Join

```
SELECT *
FROM class
CROSS JOIN class_info;
```

Output:



id	name	id	address
4	Mokshagna	1	DELHI
2	Sowmya	1	DELHI
1	Sushma	1	DELHI
4	Mokshagna	2	MUMBAI
2	Sowmya	2	MUMBAI
1	Sushma	2	MUMBAI
4	Mokshagna	3	CHENNAI
2	Sowmya	3	CHENNAI
1	Sushma	3	CHENNAI

Inner Join

```
SELECT *
FROM class
INNER JOIN class_info
ON class.id = class_info.id;
```

Output:

	id	name	id	address
▶	1	Sushma	1	DELHI
	2	Sowmya	2	MUMBAI

Inner Join

```
SELECT class.name,
       class_info.address
FROM class
INNER JOIN class_info
ON class.id=class_info.id;
```

Output:

	name	address
▶	Sushma	DELHI
	Sowmya	MUMBAI

Natural Join

```
SELECT *
FROM class
NATURAL JOIN class_info;
```

Output:

	id	name	address
▶	1	Sushma	DELHI
	2	Sowmya	MUMBAI

Left Outer Join

```
INSERT INTO class VALUES
(5,'Krishna');
INSERT INTO class_info VALUES
(7,'NOIDA'),
(8,'PANIPAT');
```

Output:

```
125  23:20:10  INSERT INTO class VALUES (5,'Krishna')
126  23:20:13  INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')
```

Left Join

```
SELECT *
FROM class
LEFT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output:

id	name	id	address
1	Sushma	1	DELHI
2	Sowmya	2	MUMBAI
4	Mokshagna	NULL	NULL
5	Krishna	NULL	NULL

Left Join

```
SELECT *
FROM class
LEFT JOIN class_info
ON class.id=class_info.id
WHERE class_info.id IS NULL;
```

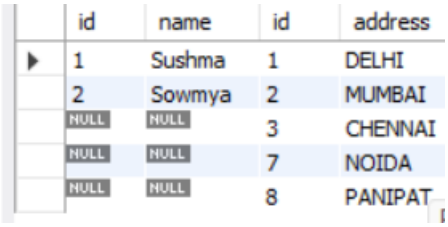
Output:

id	name	id	address
4	Mokshagna	NULL	NULL
5	Krishna	NULL	NULL

Right Join

```
SELECT *
FROM class
RIGHT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output:

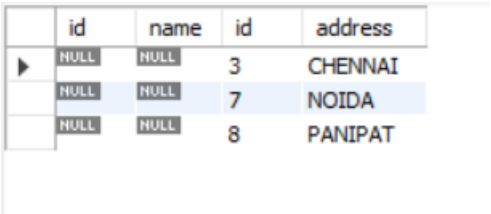


	id	name	id	address
▶	1	Sushma	1	DELHI
	2	Sowmya	2	MUMBAI
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Right Join

```
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id=class_info.id
WHERE class.id IS NULL;
```

Output:

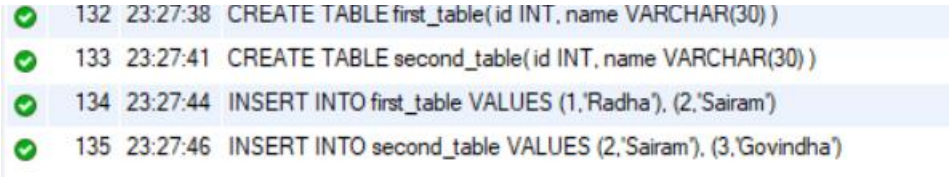


	id	name	id	address
▶	NULL	NULL	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Union

```
CREATE TABLE first_table(
id INT,
name VARCHAR(30)
);
CREATE TABLE second_table(
id INT,
name VARCHAR(30)
);
INSERT INTO first_table VALUES
(1,'Radha'),
(2,'Sairam');
INSERT INTO second_table VALUES
(2,'Sairam'),
(3,'Govindha');
```

Output:



✓	132	23:27:38	CREATE TABLE first_table(id INT, name VARCHAR(30))
✓	133	23:27:41	CREATE TABLE second_table(id INT, name VARCHAR(30))
✓	134	23:27:44	INSERT INTO first_table VALUES (1,'Radha'), (2,'Sairam')
✓	135	23:27:46	INSERT INTO second_table VALUES (2,'Sairam'), (3,'Govindha')

Union:

```
SELECT * FROM first_table
UNION
SELECT * FROM second_table;
```

Output:

	id	name
▶	1	Radha
	2	Sairam
	3	Govindha

Union All

```
SELECT * FROM first_table
UNION ALL
SELECT * FROM second_table;
```

Output:

	id	name
▶	1	Radha
	2	Sairam
	2	Sairam
	3	Govindha

Intersect

```
SELECT * FROM first_table
INTERSECT
SELECT * FROM second_table;
```

Output:

	id	name
▶	2	Sairam

Minus

```
SELECT first_table.*
FROM first_table
LEFT JOIN  second_table
ON  first_table.id=second_table.id
WHERE second_table.id IS NULL;
```

Output:

	id	name
▶	1	Radha

Address availability status:

```
SELECT c.id,
       c.name,
       CASE
           WHEN ci.address IS NULL
           THEN 'Address Missing'
           ELSE 'Address Available'
       END AS Status
FROM class c
LEFT JOIN class_info ci

ON c.id=ci.id;
```

Output:

	id	name	Status
▶	1	Sushma	Address Available
	2	Sowmya	Address Available
	4	Mokshagna	Address Missing
	5	Krishna	Address Missing